

Monitoring and Understanding Notebook-Based Learning with Jupyter

Universität
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MUC Student Research
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Problem

Final notebook submissions
hide the learning process.



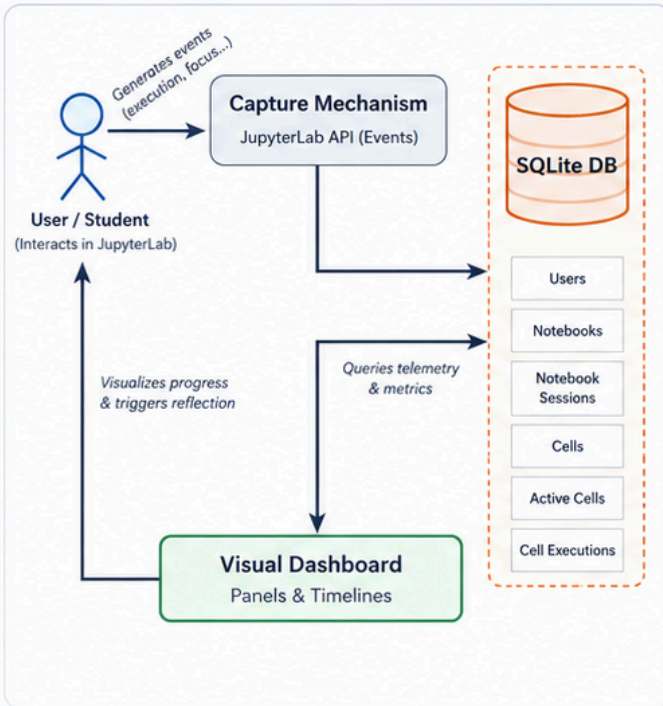
Our Approach

Capture fine-grained JupyterLab events
and compute real-time analytics.



Goal

Support student reflection with
live cell-level insights.



NAMA1968

ACCESS LEVEL: Student

SUCCESS/ERROR/ATTEMPT RATE ?



ERROR RATE ?



33%

TOP ERROR CELLS ?

Limit: 5

Cell In [15]
Given tuple
numbers = (9, 3, 9, 9, 6, 9, 2 ... 4 fails

Cell In [8]
Write your code for exercise
...
l p ... 1 fail

TOP ATTEMPT CELLS ?

Limit: 2

Cell In [9]
Write your code for exercise ... 7 attempts

Cell In [15]
Given tuple
numbers = (9, 3, 9, 9, 6, 9, 2 ... 4 attempts

TOP TIME-CONSUMING CELLS ?

Limit: 2

Cell In [15]
Given tuple
numbers = (9, 3, 9, 9, 6, 9, 2 ... 10m 24s

Cell In [11]
Create sets for Amelia and B
... 10s

TOP COMMON ERROR TYPES ?

Limit: 5

IndexError	3
SyntaxError	1
NameError	1

EXECUTION TIMELINE ?

